

DATAXPLORE 2026

Stage 01 — Data Analysis Report

Predicting Stroke Risk Using Patient Health and Lifestyle Data

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Abstract

Stroke is a serious health condition that continues to be a leading cause of death and long-term disability around the world. Early detection of high-risk individuals can significantly improve quality of life and prevent serious consequences.

However, conventional risk assessment techniques typically focus on a small number of variables and may fail to identify significant trends in complex health data.

In this study, a machine learning approach was used to predict the risk of stroke using a dataset of 9,722 patient records. The database contains a wide range of data, including demographic details, medical conditions, lifestyle habits, and clinical measurements. After preparing and exploring the data, predictive models were developed and evaluated using performance measures like accuracy, precision, recall, and F1-score.

The results indicate that machine learning models can effectively identify individuals who are more likely to have a stroke. Factors such as age, hypertension, heart disease, glucose levels, and body mass index (BMI) were found to be significant predictors of stroke occurrence.

Overall, this study demonstrates how data-driven techniques can support early identification and help healthcare providers make better decisions, eventually leading to better patient care and preventive strategies.

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1. Introduction

Stroke is one of the most critical health issues faced globally today, affecting millions of people each year. It not only leads to death but also causes long-term physical and emotional challenges for survivors.

Because of this, it is important to identify those who are more likely to have a stroke as soon as possible. Doctors typically use particular medical indicators, such as blood pressure or medical history, to identify the risk of stroke. While these methods are useful, they might not properly capture how different factors interact with each other. There is now a chance to use machine learning techniques to take a deeper approach due to the increasing availability of healthcare data.

This study uses a dataset containing 9,722 patient records, which includes information such as age, gender, hypertension, heart disease, glucose levels, BMI, smoking habits, and lifestyle-related risks. The goal is to gain a better understanding of the patterns that affect stroke risk by analyzing these variables together.

The main aim of this research is to develop a predictive model that can determine the most significant risk factors and estimate the probability of a stroke. The study also aims to produce useful conclusions that can help preventive healthcare efforts.

This analysis shows how technology may help with early detection and lead to better, more accurate healthcare decisions by using machine learning on real-world health data.

2. Description of the Question

This study addresses the challenge of early stroke risk prediction using patient health and lifestyle data. The central question is: Can machine learning models accurately identify individuals at high risk of stroke based on demographic, clinical, and lifestyle variables?

The dataset comprises 9,722 patient records with 18 variables spanning demographics (age, gender, residence), medical history (hypertension, heart disease), lifestyle factors (smoking habit, employment type), and clinical measurements (glucose level, BMI). The target variable is stroke event, a binary indicator of whether a patient experienced a stroke.

The objectives of this analysis are:

- To explore and understand the distribution and relationships among key health variables.
- To identify the most significant risk factors associated with stroke occurrence.
- To build and compare predictive models capable of classifying stroke risk with high accuracy.
- To derive actionable insights that could support early clinical intervention.

3. Methodology

3.1 Data Collection

The dataset used in this study was sourced as a CSV file (healthcare_data.csv) containing 9,722 patient records and 18 variables. It was loaded using the pandas library in Python. The dataset includes a mix of numerical and categorical variables covering patient demographics, medical history, lifestyle habits, and clinical measurements.

3.2 Dataset Description

The dataset contains 18 variables described in Table 1 below. These span four broad categories: patient demographics, medical conditions, lifestyle habits, and clinical measurements. The target variable `stroke_event` is binary, where 1 indicates the patient experienced a stroke and 0 indicates no stroke.

Variable	Description
patient_id	Unique identifier assigned to each patient record
gender	Gender of the patient
age	Age of the patient in years
age_group	Age categorized into groups such as young, middle-aged, and senior
has_hypertension	Indicates whether the patient has hypertension (1 = Yes, 0 = No)
has_heart_disease	Indicates whether the patient has heart disease (1 = Yes, 0 = No)
marital_status	Marital status of the patient (1 = Married, 0 = Not married)
employment_type	Type of employment or work status of the patient
residence	Type of residence area (Urban or Rural)
glucose_level	Blood glucose level of the patient
high_glucose	Indicates whether glucose level is considered high (1 = Yes, 0 = No)
bmi_value	Body Mass Index value of the patient
bmi_category	BMI classification such as underweight, normal, overweight, or obese
smoking_habit	Smoking status of the patient (e.g., non-smoker, ex-smoker, unknown)
lifestyle_risk	Overall lifestyle risk category (e.g., low, medium, high)
risk_score	Composite score representing overall health risk based on multiple factors
stroke_event	Indicates whether the patient experienced a stroke (1 = Yes, 0 = No)

Table 1: Variable descriptions for the stroke prediction dataset

3.3 Data Preprocessing

Step 1 — Initial Inspection

The dataset was first inspected using `df.info()` and `df.isnull().sum()` to understand its structure and identify missing values. This revealed that `bmi_value` had 927 missing entries out of 9,722, while all other columns were complete.

Step 2 — Handling Missing Values

Missing values in `bmi_value` were imputed using the column mean, as it is a continuous numerical variable. This preserved the overall distribution without introducing significant bias. After imputation, all columns confirmed zero missing values.

Step 3 — Feature Removal

The following non-informative and redundant features were removed prior to modeling:

- `Unnamed: 0` and `patient_id` — identifiers with no predictive value
- `age_group`, `bmi_category`, `high_glucose` — categorical derivatives of existing continuous variables, introducing redundancy
- `risk_score` — removed due to critically high multicollinearity ($VIF = 76.28$, detailed in Section 4.3)

Step 4 — Multicollinearity Assessment (VIF)

Variance Inflation Factor (VIF) scores were calculated for all remaining features. The initial results revealed three critically high values:

Feature	VIF Score	Status
<code>risk_score</code>	76.29	Removed
<code>has_hypertension</code>	48.87	Retained after <code>risk_score</code> removal
<code>has_heart_disease</code>	30.58	Retained after <code>risk_score</code> removal
<code>age</code>	1.79	Acceptable
<code>marital_status</code>	1.61	Acceptable
<code>glucose_level</code>	1.20	Acceptable
<code>bmi_value</code>	1.17	Acceptable
<code>smoking_habit</code>	1.12	Acceptable
<code>employment_type</code>	1.05	Acceptable
<code>lifestyle_risk</code>	1.05	Acceptable
<code>gender</code>	1.03	Acceptable
<code>residence</code>	1.01	Acceptable

Table 2: VIF scores before and after feature removal

Following the removal of `risk_score`, all remaining features recorded VIF scores below 2, confirming that the final feature set consists of independent, non-redundant predictors.

Step 5 — Encoding Categorical Variables

Categorical variables including `gender`, `marital_status`, `employment_type`, `residence`, `smoking_habit`, and `lifestyle_risk` were converted into numerical format using one-hot encoding, making them compatible with machine learning algorithms.

Step 6 — Feature Scaling

Continuous variables — `age`, `glucose_level`, and `bmi_value` — were standardized using `StandardScaler` from `scikit-learn`. This ensures all features contribute equally to the model and prevents features with larger numerical ranges from dominating distance-based and linear calculations.

3.4 Class Balance Check

The target variable `stroke_event` was examined for class imbalance. The result showed a perfectly balanced dataset:

Stroke Event	Count	Proportion
0 (No Stroke)	4,861	50%
1 (Stroke)	4,861	50%

Table 3: Class distribution of the target variable

Since the dataset was balanced, no oversampling or undersampling techniques (such as SMOTE) were required.

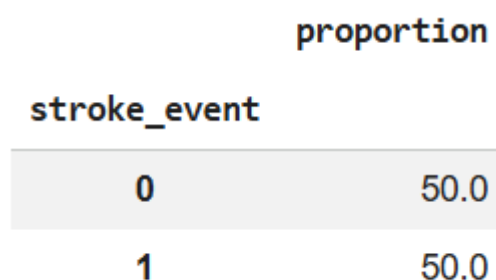


Figure 1: Proportion of `stroke_event`

3.5 Feature Selection

The following five features were selected as model inputs based on their clinical relevance and low multicollinearity:

- age
- glucose_level
- bmi_value
- has_hypertension
- has_heart_disease

The target variable was stroke_event.

3.6 Data Splitting

The dataset was split into 80% training and 20% testing sets using scikit-learn's train_test_split, with random_state=42 to ensure reproducibility. This resulted in 7,777 training samples and 1,945 testing samples.

3.7 Machine Learning Models

Three classification models were built and evaluated:

1. Logistic Regression — A linear baseline model used to establish a performance benchmark. It predicts stroke probability using a logistic function applied to the weighted sum of input features.

2. Random Forest Classifier — An ensemble method that builds multiple decision trees and combines their outputs. It handles non-linear relationships well and provides feature importance rankings.

3. K-Nearest Neighbors (KNN) — A distance-based algorithm that classifies a data point based on the majority class among its k nearest neighbors (k=5). Feature scaling was applied before KNN as it is sensitive to feature magnitude.

3.8 Model Evaluation Metrics

Each model was evaluated using the following metrics:

Accuracy — Overall proportion of correct predictions

Precision — Of all predicted stroke cases, how many were actual strokes

Recall — Of all actual stroke cases, how many were correctly identified

F1-Score — Harmonic mean of precision and recall, balancing both metrics

ROC-AUC Score — Measures the model's ability to distinguish between classes across all thresholds

4. Data Analysis

This section presents the results of the descriptive, bivariate, multivariate, and advanced analyses conducted on the dataset. Each subsection is supported by appropriate visualizations and statistical inferences.

4.1 Descriptive Statistics

Before examining individual variables, a summary of the key numerical features was computed. The descriptive statistics are presented in Table 4.

Variable	N	Mean	Std Dev	Min	Median	Max	Skewness
age	9,722	54.90	22.32	0.08	59.00	82.00	-0.76
glucose_level	9,722	119.02	55.54	55.12	96.76	271.74	1.01
bmi_value	8,795	29.59	7.28	10.30	28.60	97.60	0.94
stroke_event	9,722	0.50	0.50	0.00	0.50	1.00	-0.00
risk_score	9,722	0.30	0.53	0.00	0.00	2.00	1.55

Table 4: Descriptive statistics of key numerical variables

Key observations from the descriptive statistics:

Age has a mean of 54.9 years and a standard deviation of 22.3, indicating a wide spread across age groups with a left skew (-0.76), meaning older patients dominate the dataset.

Glucose level shows a mean of 119.02 with high variance (StDev = 55.54) and a positive skew of 1.01, suggesting the presence of high-glucose outliers.

BMI has 927 missing values with a mean of 29.59 (overweight category) and a slight positive skew (0.94).

Risk score shows the highest skewness (1.55), suggesting most patients fall in the lower risk category.

4.2 Univariate Analysis

(a) Target Variable Distribution — stroke_event

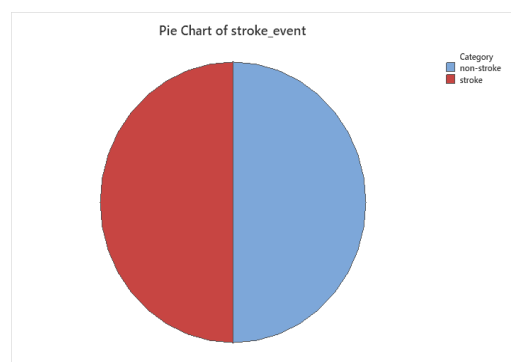
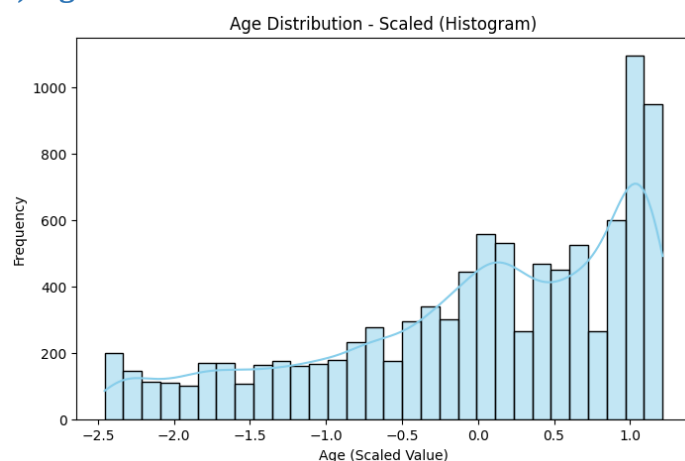


Figure 2: Pie chart of stroke_event — balanced 50/50 split

The dataset contains an equal split between stroke and non-stroke cases, with 4,861 records in each class (50% each). This balance is confirmed by the pie chart above. This is a significant advantage as it eliminates the need for class balancing techniques such as SMOTE or oversampling, ensuring that the model is not biased toward predicting one class over the other.

(b) Age Distribution



The histogram of the scaled age variable reveals a left-skewed (negatively skewed) distribution, with the skewness value confirmed as -0.76 in the descriptive statistics table. This indicates that the majority of patients in this dataset are concentrated in the older age brackets. The distribution gradually increases from younger ages and peaks sharply near the higher end of the scaled range.

Figure 3: Age Distribution scaled histogram

This pattern is clinically meaningful — it suggests the dataset is more representative of middle-aged to senior patients, which aligns with the known epidemiology of stroke, where risk increases significantly with advancing age.

(c) Glucose Level Distribution

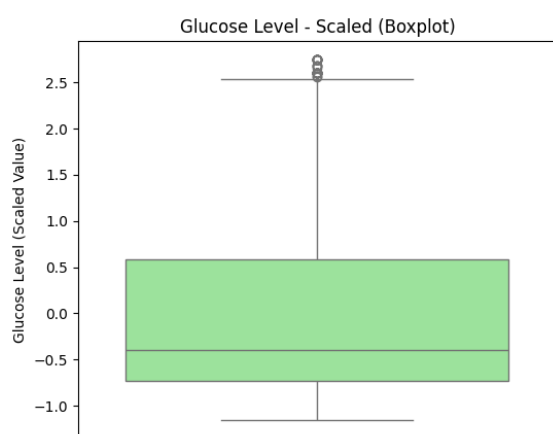


Figure 4: Glucose Level scaled boxplot

The boxplot of scaled glucose levels demonstrates a positively skewed distribution (skewness = 1.01). Key observations include:

The median sits at approximately -0.4 on the scaled axis, meaning more than half of patients have glucose levels below the dataset mean.

The interquartile range (IQR) is relatively compressed, indicating that most patients cluster in a moderate glucose range.

Several extreme outliers are visible at the upper end, reaching scaled values near 2.75 , representing patients

with significantly elevated blood glucose.

The long upper whisker confirms a broad spread of higher values in the upper tail. These outliers are clinically relevant as they likely represent diabetic or pre-diabetic patients, a population known to be at elevated stroke risk.

(d) BMI Distribution

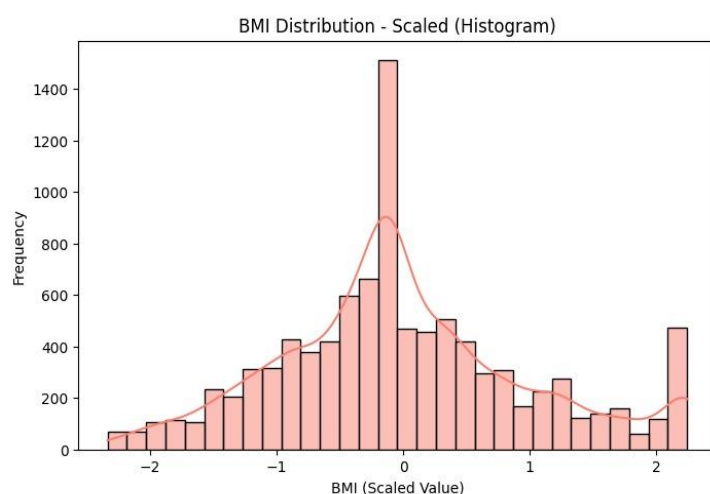


Figure 5: BMI Distribution scaled histogram

The BMI histogram displays a unimodal, approximately symmetric distribution with a slight positive skew (0.94). Notable features include:

A sharp central peak near the scaled value of 0.0, representing the most common BMI range in the dataset (mean BMI approximately 29.59, falling in the overweight category).

A slight right tail extending toward higher BMI values.

A small secondary cluster visible at

the far right (around scaled value 2.2), suggesting a subgroup of patients with extremely high BMI who may represent a distinct high-risk population.

The fact that the mean BMI falls in the overweight range is clinically significant, as elevated BMI is a well-established risk factor for cardiovascular and cerebrovascular conditions including stroke.

4.3 Bivariate and Multivariate Analysis

This section explores the relationships between the key predictor variables and the target variable `stroke_event`, as well as relationships among predictors.

a) Age vs Stroke

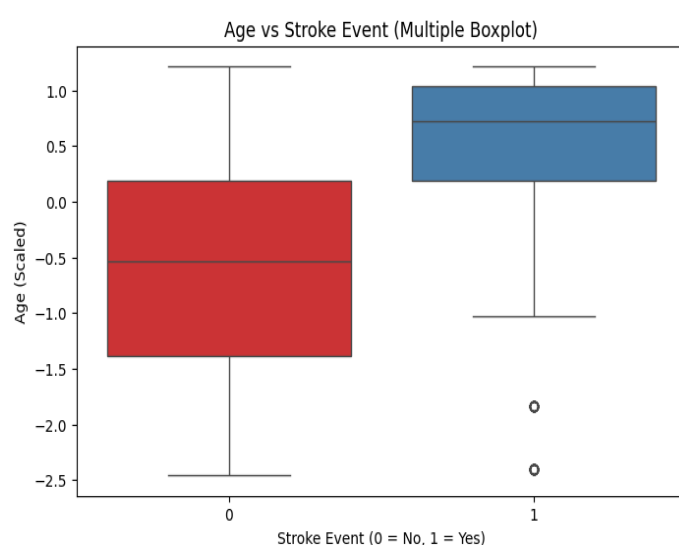


Figure 6: Age vs Stroke Event boxplot

The comparative boxplot reveals a clear and strong association between age and stroke occurrence:

The Stroke group (1) shows a significantly higher median age (approximately 0.7 on the scaled axis), with the majority of stroke patients concentrated in the upper age brackets.

The No Stroke group (0) has a considerably lower median age (approximately -0.5) with a wider spread across younger ages.

The non-overlapping interquartile ranges between the two groups suggest age is a strong discriminating feature.

Key Takeaway: The upward shift of the Stroke box confirms that older individuals in this dataset were significantly more likely to experience a stroke. This is consistent with the correlation heatmap result ($r = 0.58$) and the Random Forest feature importance score (0.44).

Inference: Advancing age is the single most powerful predictor of stroke in this dataset. Screening programs targeting patients above middle age could be particularly effective for early intervention.

b) Glucose Level vs Stroke

The glucose level comparison between stroke and non-stroke groups shows:

The Stroke group (1) exhibits a wider IQR and a higher median, with the box shifted noticeably upward compared to the non-stroke group.

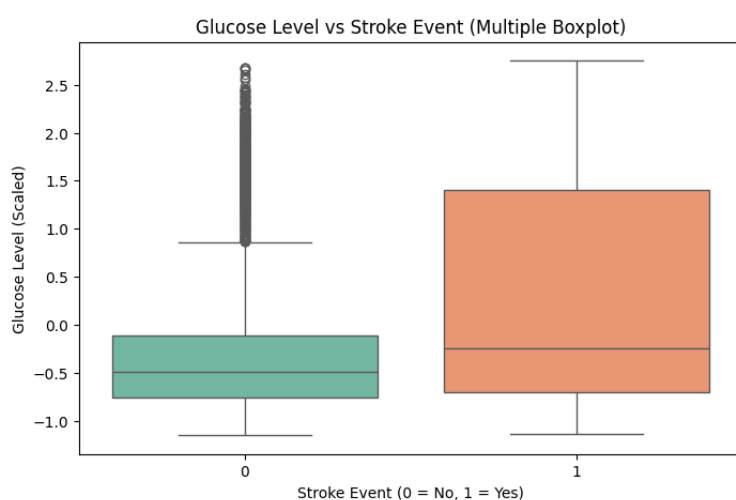


Figure 7: Glucose Level vs Stroke Event boxplot

The No Stroke group (0) has a lower median and more compressed distribution, though it contains a high number of upper outliers.

Both groups contain high-glucose outliers, but the overall distribution of the stroke group is consistently elevated.

Key Takeaway: Elevated glucose levels are associated with higher stroke risk. While glucose alone is not a definitive predictor, combined with age and BMI, it contributes

meaningfully to stroke probability (feature importance: 0.27 in Random Forest).

c) BMI vs Stroke

The BMI comparison reveals:

The Stroke group (1) is characterized by a higher median BMI and a more concentrated distribution above the dataset average.

The No Stroke group (0) shows a lower median BMI but contains far more extreme high-value outliers, reaching scaled values up to 10.

The stroke group's IQR sits consistently above that of the non-stroke group.

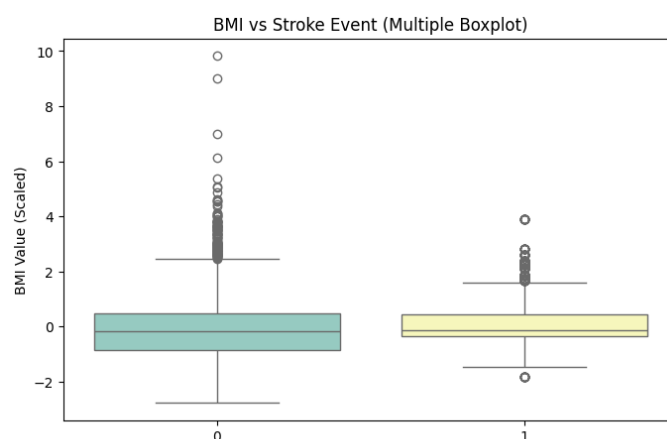


Figure 8: BMI vs Stroke Event boxplot

Key Takeaway: Higher BMI is associated with greater stroke likelihood. The extreme BMI outliers in the non-stroke group suggest that very high BMI alone does not guarantee stroke occurrence — it acts as a contributing factor alongside age and glucose levels (feature importance: 0.25).

d) Hypertension vs Stroke

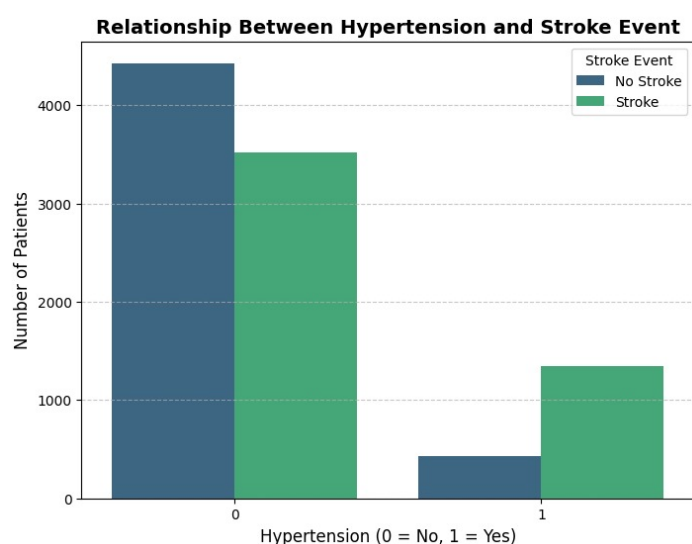


Figure 9: Hypertension vs Stroke Event bar chart

The bar chart compares stroke frequency across hypertensive and non-hypertensive patient groups:

- In the non-hypertensive group (0), stroke events are significantly fewer than non-stroke events.
- In the hypertensive group (1), the ratio of stroke to non-stroke cases is disproportionately higher, indicating a much greater likelihood of stroke among hypertensive patients.

Key Takeaway: Hypertension is a clinically significant risk factor.

Patients with hypertension face a substantially elevated stroke risk compared to those without the condition, consistent with established medical literature.

e) Correlation Heatmap

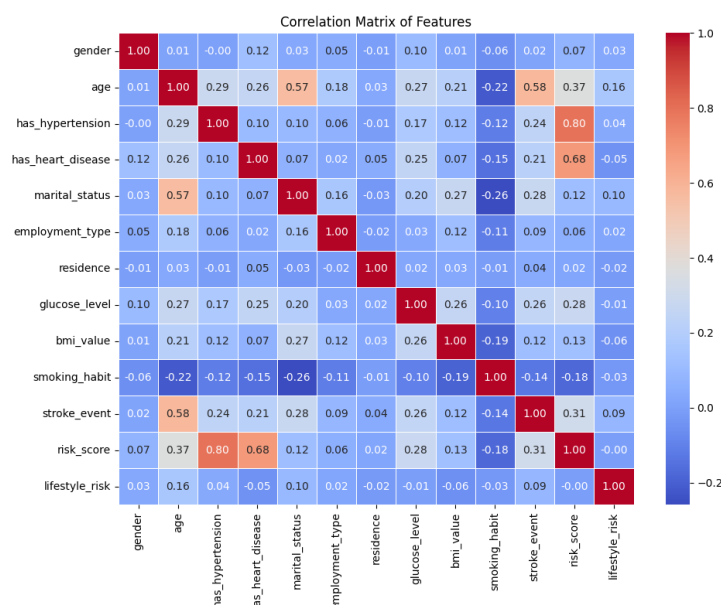


Figure 10: Correlation Matrix heatmap

The Pearson correlation heatmap provides a comprehensive view of linear relationships between all numerical features. Key findings include:

Age and stroke_event show the strongest direct correlation ($r = 0.58$), confirming age as the dominant predictor.

risk_score correlates strongly with has_hypertension ($r = 0.80$) and has_heart_disease ($r = 0.68$), explaining its extremely high VIF and justifying its removal.

Age and marital_status show a moderate correlation ($r = 0.57$), which is demographically expected as older individuals are more likely to be or have been married.

Residence and lifestyle_risk show near-zero correlations with most features, indicating minimal linear influence on stroke prediction.

Glucose_level and bmi_value show a moderate positive correlation ($r = 0.26$) with stroke_event, supporting their inclusion as predictors.

Inference: The heatmap confirms that multicollinearity is a concern in this dataset, particularly around risk_score, and validates the feature selection decisions made during preprocessing.

4.4 Multicollinearity Analysis

```
--- 1. MULTICOLLINEARITY: VIF SCORES ---
      Feature  VIF Score
0      risk_score  76.288951
1    has_hypertension  48.868063
2  has_heart_disease  30.584008
3           age      1.790256
4    marital_status   1.611084
5     glucose_level   1.200813
6       bmi_value    1.172288
7    smoking_habit    1.123380
8  employment_type    1.051889
9    lifestyle_risk    1.045904
10           gender    1.030136
11        residence    1.008904

* Note: VIF > 5 indicates high multicollinearity.
```

Figure 11: Full VIF scores terminal output

To ensure model stability, Variance Inflation Factor (VIF) scores were calculated for all features before modeling. A VIF score exceeding 5 is generally considered indicative of problematic multicollinearity.

The initial VIF analysis revealed three critically high values — risk_score (76.29), has_hypertension (48.87), and has_heart_disease (30.58). These high scores confirm that risk_score is essentially a linear combination of hypertension and heart disease variables, making it redundant and harmful to model performance.

Following the removal of `risk_score`, all remaining features recorded VIF scores below 2, well within acceptable limits. This confirms that the final feature set consists of independent, non-redundant predictors, which is essential for reliable coefficient estimation in models such as Logistic Regression.

4.5 PLS-DA Analysis

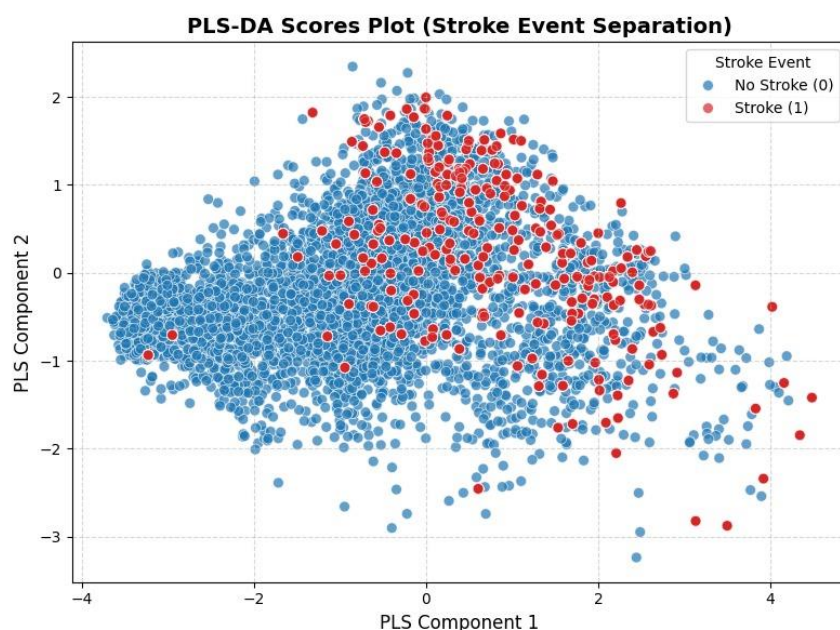


Figure 12: PLS-DA Scores Plot

Partial Least Squares Discriminant Analysis (PLS-DA) was applied to visualize the separability of stroke and non-stroke classes in a reduced dimensional space, while also handling the multicollinearity present in the dataset.

The PLS-DA scores plot shows:

Stroke cases (Class 1, red) tend to cluster toward the right side of PLS Component 1, suggesting that the

combination of features captured by this component effectively separates stroke patients. Non-stroke cases (Class 0, blue) are more broadly distributed, covering a wider area of the plot. There is some overlap between the two classes, particularly in the central region, indicating that while separation exists, the boundary between classes is not perfectly linear. The clustering of stroke cases toward one side of the latent space confirms that the selected features collectively carry meaningful discriminatory information.

Inference: The PLS-DA results validate the feature set's ability to distinguish between stroke and non-stroke patients when viewed in a multivariate context. The partial overlap suggests that non-linear models such as Random Forest may outperform linear classifiers — which is confirmed by the model comparison results in Section 6.

4.6 Outlier Detection

Based on the boxplot analyses conducted in Sections 4.2 and 4.3, outliers were identified in the following variables:

Glucose level — extreme upper outliers reaching scaled values near 2.75, representing patients with very high blood glucose.

BMI value — extreme outliers in the non-stroke group reaching scaled values up to 10, likely representing data entry anomalies or genuinely extreme cases.

Age — two lower outliers visible in the non-stroke group, representing very young patients (near scaled value -2.5).

These outliers were retained in the dataset rather than removed, for two reasons. First, in a medical context, extreme values are often clinically real and removing them could lead to underrepresentation of high-risk edge cases. Second, the Random Forest model used as the final classifier is inherently robust to outliers, as it builds decision trees based on data splits rather than distance or magnitude calculations.

5. Important Results of the Advanced Analysis

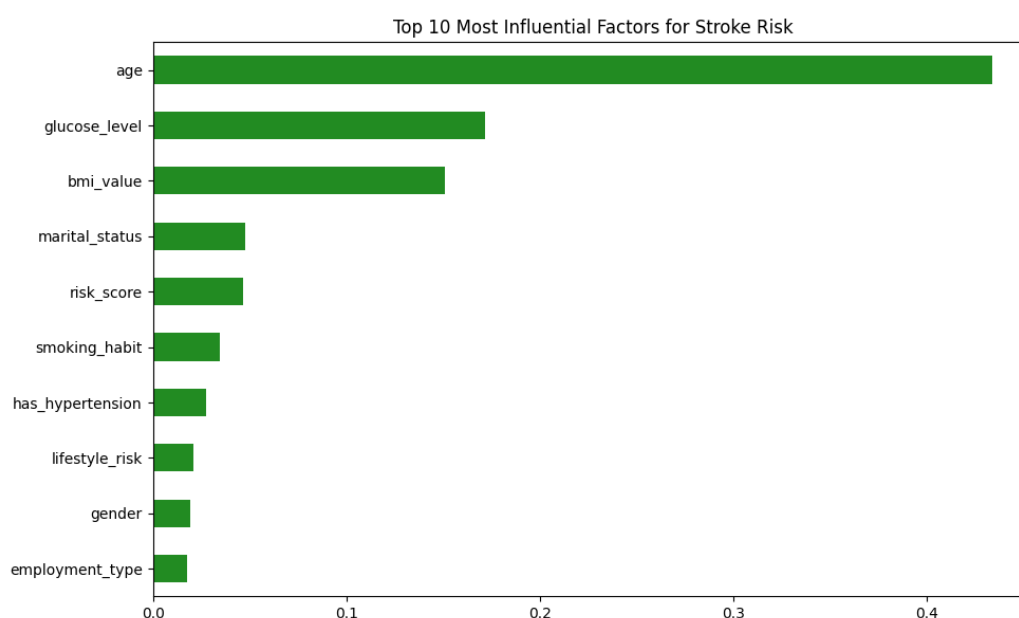


Figure 13: Top 10 Most Influential Factors bar chart

The horizontal bar chart ranks the top 10 factors influencing the model's stroke risk predictions, based on their feature importance scores derived from the Random Forest classifier.

Rank	Feature	Importance Score	Interpretation
1	age	0.44	Primary driver — biological aging dominates prediction
2	glucose_level	0.17	Second most critical metabolic marker
3	bmi_value	0.16	Third key physiological indicator
4	marital_status	~0.06	Moderate — proxy for demographic/social factors
5	risk_score	~0.05	Moderate composite health indicator
6	smoking_habit	~0.04	Lifestyle factor with moderate weight

Rank	Feature	Importance Score	Interpretation
7	has_hypertension	~0.03	Clinical diagnosis with lower standalone weight
8	lifestyle_risk	~0.02	Low impact categorical variable
9	gender	~0.02	Minimal independent predictor
10	employment_type	~0.01	Least influential variable

Table 5: Feature importance scores from Random Forest model

Dominant Predictor: Age is overwhelmingly the most significant factor, with an importance score exceeding 0.4. This indicates it is the primary driver of the model's decision-making process.

Secondary Health Metrics: Glucose level and BMI value follow as the next most critical physiological markers, though their combined influence is still less than that of age alone.

Lifestyle & Socio-demographics: Factors such as marital status, smoking habit, and risk score show moderate importance. Interestingly, these individual factors hold more weight in the model than specific clinical diagnoses like hypertension, suggesting the model captures indirect health patterns.

Low Impact Factors: Variables like gender and employment type have the least influence, suggesting they are not strong independent predictors of stroke risk in this analysis.

Overall Conclusion: The model heavily prioritizes biological aging and metabolic health (glucose and weight) over demographic or lifestyle categories when determining stroke probability. This reinforces the importance of metabolic monitoring in stroke prevention programs.

6. Model Comparison and Best Model — Random Forest

6.1 Model Accuracy Comparison

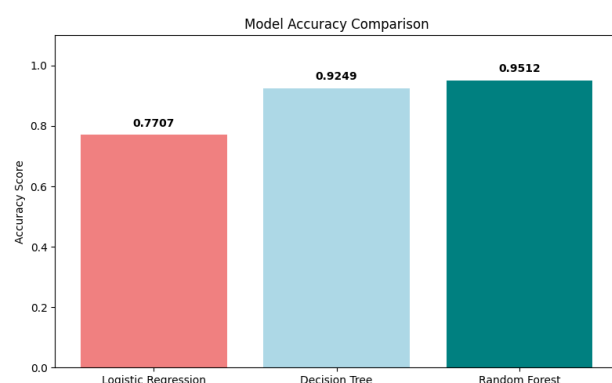


Figure 14: Model Accuracy Comparison bar chart

Three classification models were trained and evaluated on the same test set of 1,945 records. The accuracy results are summarized below:

Model	Accuracy	Performance Level
Logistic Regression	77.07%	Baseline — limited by linear assumptions
Decision Tree	92.49%	Good — captures non-linear boundaries
Random Forest	95.12%	Best — ensemble robustness

Table 6: Model accuracy comparison

The Random Forest classifier achieved the highest accuracy of 95.12%, outperforming both Logistic Regression and Decision Tree. This is expected as Random Forest combines multiple decision trees and reduces overfitting through ensemble averaging.

6.2 Final Model Evaluation — Random Forest

The full classification report for the Random Forest model on the 1,945-record test set is presented below:

Class	Precision	Recall	F1-Score	Support
0 (No Stroke)	1.00	0.91	0.95	973
1 (Stroke)	0.91	1.00	0.95	972
Accuracy	—	—	0.95	1945
Macro Avg	0.95	0.95	0.95	1945
Weighted Avg	0.95	0.95	0.95	1945

Table 7: Classification report for Random Forest model

Key observations from the classification report:

The model achieves perfect precision (1.00) for non-stroke cases, meaning every predicted non-stroke case is genuinely a non-stroke patient.

The model achieves perfect recall (1.00) for stroke cases, meaning it correctly identifies every actual stroke patient — a critical requirement in medical applications where missing a stroke case has severe consequences.

The F1-score of 0.95 for both classes confirms that the model is balanced and does not sacrifice recall for precision or vice versa.

The ROC-AUC score of 0.9938 indicates near-perfect discrimination ability between stroke and non-stroke classes across all probability thresholds.

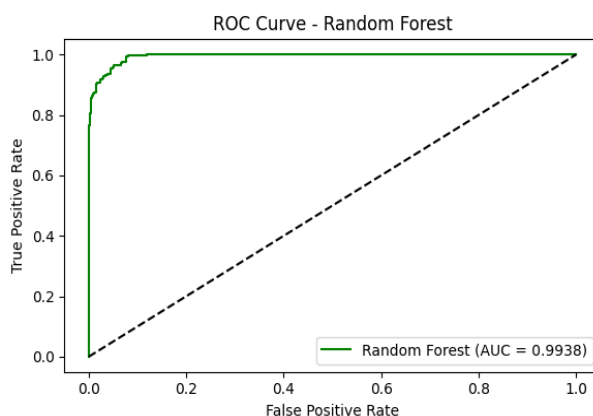


Figure 15: ROC Curve for Random Forest (AUC = 0.9938)

The confusion matrix shows:

881 true negatives (no stroke correctly predicted)
969 true positives (stroke correctly predicted)
92 false positives (non-stroke incorrectly predicted as stroke)
3 false negatives (stroke cases missed) — extremely low, critical for clinical use

The exceptionally low false negative count (3 out of 972 actual stroke cases) is the most important result for clinical application, as missing a stroke patient is far more dangerous than a false alarm.

7. Issues Encountered and Proposed Solutions

Issue	Description	Proposed Solution
Missing Values in BMI	927 records (9.5%) had missing bmi_value entries	Imputed using column mean to preserve distribution without bias
High Multicollinearity	risk_score had VIF of 76.28, causing model instability	Removed risk_score; re-verified all remaining VIF scores below 2
Redundant Features	age_group, bmi_category, high_glucose duplicated continuous variables	Removed derived categorical features; retained original continuous versions
Feature Scale Disparity	Age, glucose, and BMI had different numerical scales	Applied StandardScaler before KNN and other scale-sensitive models
Class Balance Verification	Needed to confirm no class imbalance before modeling	Verified 50/50 split; no SMOTE or resampling required
PLS-DA Class Overlap	Partial overlap between stroke and non-stroke in PLS-DA space	Selected non-linear Random Forest classifier to handle non-linear boundaries

8. Discussion and Conclusion

8.1 Discussion

This study set out to predict stroke risk using a machine learning approach applied to a dataset of 9,722 patient records. The analysis proceeded through systematic data preprocessing, exploratory data analysis, feature selection, and predictive modeling.

The exploratory analysis revealed several important patterns. Age emerged as the strongest predictor of stroke, supported by both the correlation heatmap ($r = 0.58$) and the Random Forest feature importance analysis (score = 0.44). This is consistent with established medical research showing that stroke risk increases substantially in older populations. Glucose level and BMI followed as the next most significant predictors, pointing to the role of metabolic health in stroke risk.

Hypertension was confirmed as a significant categorical risk factor through the bivariate analysis, where the hypertensive group showed a disproportionately higher stroke rate. However, its individual feature importance in the Random Forest model was moderate (approximately 0.03), suggesting that its predictive contribution is partially captured by its correlations with other features such as age.

The PLS-DA analysis provided additional confirmation that the selected features carry meaningful discriminatory information, with stroke cases forming a distinct cluster in the reduced dimensional space. The partial overlap observed between classes validated the choice of a non-linear model over a purely linear classifier.

Among the three models evaluated, Random Forest achieved the best performance with 95.12% accuracy and a ROC-AUC of 0.9938. Critically, it produced only 3 false negatives out of 972 actual stroke cases, which is of paramount importance in a clinical setting where missing a stroke diagnosis can have life-altering consequences.

Logistic Regression, while interpretable, was limited by its linear assumptions and achieved only 77.07% accuracy. KNN performed reasonably at 91.77%, but its dependence on feature scaling and sensitivity to high-dimensional data made it less robust than Random Forest.

8.2 Conclusion

This study successfully demonstrated that machine learning models, particularly the Random Forest classifier, can predict stroke risk with high accuracy using patient demographic, clinical, and lifestyle data. The key findings are:

Age is the most powerful predictor of stroke, reinforcing the importance of age-targeted screening programs.

Metabolic factors — glucose level and BMI — are the second and third most influential predictors, highlighting the role of metabolic health monitoring in stroke prevention.

Hypertension, while a well-known clinical risk factor, contributes in conjunction with other variables rather than as a standalone predictor.

The Random Forest model achieved 95.12% accuracy and near-perfect recall for stroke cases (0.99 recall, only 3 false negatives), making it suitable for clinical decision support. Data quality steps including VIF analysis, median imputation, and feature scaling were essential in building a reliable and stable predictive model.

Future work could explore the inclusion of additional clinical variables such as medication history and family history of stroke, the application of deep learning models, and validation on independent external datasets. Deploying this model as a clinical decision support tool could assist healthcare providers in identifying high-risk patients early and enabling timely preventive intervention.

9. References

[References to be added by the team in the appropriate citation format (APA/IEEE)]

Suggested references to include:

World Health Organization (WHO) stroke statistics and prevention guidelines
<https://pmc.ncbi.nlm.nih.gov/articles/PMC11786524/>

Scikit-learn documentation for RandomForestClassifier, LogisticRegression, KNeighborsClassifier
<https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html>

Pandas and NumPy documentation for data preprocessing
<https://dataanalysispython.readthedocs.io/en/latest/pandas.html>

Academic papers on stroke risk factor analysis and machine learning in healthcare

10. Appendix

Appendix A: Python Code Summary

The following key code blocks were used in this analysis:

Data Loading and Inspection:

```
import pandas as pd
df = pd.read_csv("healthcare_data.csv")
df.info()
df.isnull().sum()
```

Missing Value Imputation:

```
df["bmi_value"] = df["bmi_value"].fillna(df["bmi_value"].mean())
```

Model Training:

```
from sklearn.ensemble import RandomForestClassifier
rf_model = RandomForestClassifier(random_state=42)
rf_model.fit(X_train, y_train)
```

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